Thakaa at StanceEval-2026: Inference-Time Target Grounding for Arabic Stance Detection

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Abstract

StanceEval-2026 tests Arabic stance classification under target shift, with evaluation targets absent from the labelled task data. We test two ways to supply target information at inference time. For Track 1, we retrieve eight labelled Women Driving demonstrations from the external ArabicStanceX corpus and score the three label tokens with Gemma-4-31B-IT. For the unseen-target track, we prepend cached target context prompted to be factual and non-evaluative, then fuse label predictions and label-token scores from multiple LLMs, with final assignments constrained by published class totals. The systems obtain 85.22 and 91.78 F_{avg2} on Tracks 1 and 2, ranking sixth and fourth in the final standings. On held-out ArabicStanceX tweets, direct prompting scores 79.45 F_{avg2}; eight random Women Driving demonstrations raise it to 84.78, and similarity retrieval reaches 89.04. Across three target-held-out Mawqif evaluations, a fixed target card raises Gemma-4-31B-IT from 86.54 to 86.93, whereas QLoRA lowers Gemma-4-12B-IT from 84.43 to 83.77. In these separate public evaluations, same-target demonstrations and retrieval improve performance, factual target context provides small, target-dependent gains, and the tested source-only adapter does not improve.

1 Introduction

Stance detection relates a text to a specified target. Sentiment alone is insufficient when a tweet criticises an implementation but supports its goal, or praises an actor without endorsing the target. Arabic social media adds dialectal variation, sarcasm, and implicit reference (Alturayeif et al., 2022; Albalawi et al., 2026b).

StanceEval-2026 (Albalawi et al., 2026a) evaluates this relation under a target shift. Track 1 is nominally the seen-target setting, yet it tests Women Driving after training on Women Empowerment and two other topics. Its unseen-target track tests Electric Cars, a transport-technology and industrial-policy topic, and the Trimester System, a debate about Saudi school-calendar reform. The Mawqif-XT dataset was designed around these two transfer settings (Albalawi et al., 2026b).

Cross-target stance systems have represented targets explicitly (Allaway and McKeown, 2020), used contrastive objectives (Liang et al., 2022), and injected retrieved or generated knowledge (He et al., 2022; Beck et al., 2023; Zhang et al., 2024). Stance Reasoner generates explicit chain-of-thought rationales and aggregates self-consistent samples (Taranukhin et al., 2024). We instead separate a reusable target description from constrained label prediction. Arabic shared-task systems have used prompt engineering or LLM fine-tuning (Al Hariri and Abu Farha, 2024; Alghaslan and Almutairy, 2024). Unlike methods that retrieve Wikipedia passages or generate tweet-specific knowledge, our target card is a single neutral, target-only description reused for every tweet. We test this alongside retrieved same-target examples as two lightweight forms of inference-time target evidence.

Our public comparisons separately test exact-target labels, demonstration selection, factual context, and source-target adaptation. A committee analysis examines whether disagreement identifies

Track	System	F_{avg2}
1	Qwen 2.5 72B baseline	73.66
1	Retrieved demos, Gemma-4-31B-IT	**85.22**
2	Qwen 2.5 72B baseline	74.75
2	Local Gemma-4-31B-IT stack	86.58
2	GPT-5.5 target-grounded classifier	91.48
2	Full target-grounded pipeline	**91.78**

Table 1: Official test results on a 0 to 100 scale. Track 1 contains 352 Women Driving tweets; Track 2 contains 644 tweets across two unseen targets. Baseline values are reported by Albalawi et al. (2026b).

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errors, while a cross-corpus rubric study tests the transfer of annotation cues.

2 Task and data

The original Mawqif splits contain 3,502 training and 619 development tweets about COVID-19 Vaccine, Digital Transformation, and Women Empowerment. Each tweet has stance, sentiment, and sarcasm labels. Track 1 evaluates 352 Women Driving tweets. Track 2 evaluates 332 Electric Cars and 312 Trimester System tweets. The primary metric is $F_{\text{avg2}} = (F_{\text{FAVOR}} + F_{\text{AGAINST}})/2$; NONE is not averaged in its numerator (Mohammad et al., 2016). We also report F_{avg3}, the macro-F1 over all three stance labels, and overall accuracy for the final systems. Each instance pairs an Arabic tweet with a named target, and the system returns exactly one of FAVOR, AGAINST, or NONE. For example, a tweet celebrating women receiving driving licences should be classified as FAVOR when paired with the Women Driving target.

For Track 1, we use ArabicStanceX (Alkhathlan et al., 2025) as an external demonstration corpus; it is not part of the shared-task data. The corpus is publicly released under CC BY 4.0. Its 947 Women Driving tweets have no exact or Arabic-normalised text overlap with the 352 task tweets. The main public ablation uses a fixed 497-row retrieval pool and a disjoint 450-row evaluation set. A separate target-control study uses the official 756/191 train/test split. Official StanceEval inference retrieves from all 947 labelled tweets. Separate public experiments evaluate target-held-out prompting and source-only adaptation on Mawqif.

3 System overview

3.1 Track 1: retrieved in-target demonstrations

For each task tweet, we retrieve the eight nearest Women Driving examples from the full ArabicStanceX pool, following similarity-based demonstration selection (Liu et al., 2022). Similarity is cosine similarity over character-boundary TF-IDF features. We use character features to reduce sensitivity to dialectal spelling variation and short social-media text. The prompt contains the target, a factual background, the tweet, the retrieved labelled examples, and a one-sentence instruction restricting the answer to the three stance labels.

We run Gemma-4-31B-IT locally with vLLM (Kwon et al., 2023; Gemma Team, 2026). Instead of generating an explanation, the model scores the continuations FAVOR, AGAINST, and NONE, and we select the largest log-probability. The Track 1 system in Table 1 performs no parameter update and uses no class quota.

3.2 Track 2: prompted factual target context

Target descriptions. The pipeline caches two English descriptions for each unseen target. Gemini 3.6 Flash generates the card used by the GPT and Gemini classifiers, while Gemma-4-31B-IT generates the card used by its own scorer. Both prompts request two or three factual sentences about the topic in Saudi Arabia and the Arab world, explicitly forbidding opinions. Each card is generated at temperature zero and reused unchanged for every tweet about its target. The stance prompt contains the Arabic and English target names, the card, and the tweet, and requests one stance label.

Classification and arbitration. GPT-5.5 supplies the initial target-grounded label, and Gemini 3.6 Flash labels each row independently. GPT-5.5 Pro arbitrates their 44 disagreements, changing 13 initial labels. Gemma-4-31B-IT and a second GPT-5.5 pass independently score the three label tokens.

Score fusion and assignment. For row i and class c, we combine these signals with the fixed log-linear rule

$$s_{i,c} = 2\log p_{i,c}^{\text{anchor}} + \log p_{i,c}^{\text{Gemma}} + \log p_{i,c}^{\text{GPT}}. \quad (1)$$

where the anchor assigns 0.85 to its selected class and 0.075 to each alternative. For each scorer, we softmax the three label log scores and floor probabilities at 10^{-12} before taking logs. A Hungarian assignment (Kuhn, 1955) maximises the total score subject to class totals derived from the official website.[1] Its anonymous test rows give sizes and FAVOR/AGAINST/NONE percentages. Matching sizes to the released data identifies 332 tweets (43.67/35.24/21.08%) as Electric Cars and 312 (22.12/72.44/5.45%) as Trimester System. Multiplying by the respective sizes and rounding gives 145/117/70 and 69/226/17.

The Gemma scorer alone receives normalised Arabic text (NFKC, diacritic and tatweel removal, letter normalisation, and URL and mention placeholders); API classifiers receive the raw tweet. The

[1] "View Dataset Statistics" at https://stanceeval.github.io/#data; also present in the June 18, 2026 revision, before the test release.

anchor uses medium reasoning, arbitration uses high reasoning, and the score-only GPT pass disables reasoning. Fusion and count assignment complete the official pipeline; the controlled comparisons in Section 5 use public held-out labels.

4 Experimental setup

All controlled comparisons use public labels and keep evaluation rows separate from retrieval, training, and checkpoint selection. Track 1 uses the two ArabicStanceX splits described in Section 2. Track 2 compares no-card, fixed-card, and gated-card prompting under target-held-out Mawqif evaluation.

For each held-out target, the *no-card* condition supplies only the target and tweet. The *fixed-card* condition prepends the same cached factual description to every tweet about that target. The *gated-card* condition uses an independent Arabic yes/no prompt to decide whether to prepend that description for each tweet. The Gemma-4-12B-IT comparison instead contrasts the unchanged base checkpoint with a source-trained QLoRA adapter. The public card study uses Arabic Gemma-generated descriptions and one-word generation at temperature zero, not the official English-card scoring protocol.

Parameter adaptation. For parameter adaptation, we fine-tune Gemma-4-12B-IT with QLoRA (Dettmers et al., 2023). Each fold trains on two Mawqif targets and evaluates on the third. The rank-16 adapter uses scaling 32, dropout 0.05, maximum length 256, effective batch size 16, and seed 42. A 15% source-only validation slice jointly selects the learning rate and checkpoint. The grid contains 5×10^{-5} and 10^{-4}, with 2×10^{-4} also tested for the COVID-19 Vaccine fold. The selected learning rates are 10^{-4} for Women Empowerment and Digital Transformation and 5×10^{-5} for COVID-19 Vaccine; all selected checkpoints occur near two-thirds of an epoch.

5 Results

Table 1 reports the official results. The final systems also score 67.92 F_{avg3} and 81.25 accuracy on Track 1, and 83.53 and 88.82 on Track 2, pooled over both targets. The Track 1 retrieval system scores 85.22 F_{avg2}, ranking sixth among 21 teams; the complete Track 2 pipeline scores 91.78, ranking fourth among 13 teams in the final standings. Its target scores are 88.75 on Electric Cars and 95.23 on Trimester System. A local-only Track 2 comparison combines eight Gemma-4-31B-IT prompt views with a categorical stacker fitted on public Women Empowerment labels and scores 86.58. The direct GPT-5.5 classifier scores 91.48. This final-test comparison is descriptive; the 0.30-point difference is not a component ablation.

Track 1, held-out ArabicStanceX

Variant	Public	Official test
direct, no demonstrations	79.45	79.51
8 random target demos	84.78	83.85
8 retrieved target demos	**89.04**	**85.22**

Track 2, target-held-out Mawqif

Model	Variant	WE	CV	DT	Mean
Gemma-4-31B-IT	no card	87.95	92.31	**79.37**	86.54
Gemma-4-31B-IT	fixed card	**89.25**	**92.34**	79.20	**86.93**
Gemma-4-31B-IT	gated card	88.51	92.25	79.20	86.65
Gemma-4-12B-IT	base	**86.18**	**88.51**	**78.60**	**84.43**
Gemma-4-12B-IT	QLoRA	86.10	87.50	77.70	83.77

Table 2: F_{avg2} on a 0 to 100 scale. The Track 1 public column evaluates 450 ArabicStanceX tweets with a disjoint 497-row demonstration pool; the official-test column reports the same three submitted arms descriptively and is not used for component selection. WE, CV, and DT denote Women Empowerment, COVID-19 Vaccine, and Digital Transformation. Track 2 columns are separate public held-out targets. Bold marks the best result within each model block.

Target demonstrations and their selection both help. On the 450 public rows, random demonstrations improve on direct prompting, and retrieval improves further (Table 2). Exact McNemar tests on paired correctness reject equal error rates for both transitions ($p = 1.4 \times 10^{-5}$ and $p = 1.6 \times 10^{-3}$). The NONE F1 rises from 25.29 to 52.00. The official-test arms preserve the ordering but are descriptive, not an ablation. A separate 191-row public control uses fixed nine-example sets with three examples per stance, holding the corpus and annotation convention fixed: direct prompting scores 71.57; nine Women Driving examples score 81.84, compared with 73.10 for nine other-target ArabicStanceX examples (difference 8.74; 95% paired-bootstrap CI [4.40, 13.38]). Together, the comparisons distinguish access to target labels from selecting locally relevant examples.

Target cards and source-only adaptation. The fixed card's change is target-dependent: $+1.30$ on Women Empowerment, $+0.03$ on COVID-19 Vaccine, and -0.17 on Digital Transformation. The

Target	Track	Fleiss' κ
Women Driving	1	0.479
Electric Cars	2	0.479
Trimester System	2	0.711
Overall	Both	0.564

Table 3: Human stance-label agreement in Mawqif-XT, as reported by Albalawi et al. (2026b).

gated-card condition selects the card for 327 of 4,121 rows (7.9%) and reaches 86.65, below the fixed-card condition. Neither is a matched ablation of the submitted GPT-5.5 anchor. QLoRA loses on every 12B fold. This finding is specific to the tested recipe, and the 31B and 12B blocks are within-model comparisons rather than a shared-backbone comparison.

6 Error analysis

On public Women Empowerment data, a committee of GPT-5.6 Luna, GPT-5.6 Terra, and Gemma-4-31B-IT is wrong on 54 of 136 split votes (39.7%), but only 99 of 1,264 unanimous votes (7.8%). A five-fold logistic stack reaches 90.08 F_{avg2}, below the 90.38 of plain majority. Disagreement is therefore a risk marker, not a correction rule.

Qualitative inspection points to *stance attachment*. One tweet jokes that, "in the era of women's empowerment, they will not do anything to you"; the three models produce all three labels, while the annotation is NONE. Local affect is easier to recognise than whether it is directed at the named target.

A corpus-specific rubric does not transfer. Post hoc, we distilled a 60-line Arabic rubric from recurring Women Driving evaluation-corpus patterns, including sarcasm, hashtag reversal, celebration, supplication, and autonomy cues. Because its construction used inspection of that corpus, it was not part of the reported system; we test only its transfer to held-out ArabicStanceX Women Driving labels. Even after removing every verbatim fragment, the detailed rubric underperforms a four-line minimal instruction for a two-model ensemble (87.67 versus 89.18 F_{avg2}) and for both members (87.02 versus 90.11; 87.42 versus 88.04). The tested corpus-derived rubric did not improve cross-corpus prediction over the minimal instruction.

The rubric result is consistent with a difficult NONE boundary rather than missing target vocabulary alone. Table 3 shows lower human agreement for Women Driving and Electric Cars than for the Trimester System. Across the three targets, 66.12% of stance disagreements contrast NONE with a polar label. These statistics do not establish a cause, but they place the transfer failure in context: lexical cues alone may not encode a corpus's annotation policy.

7 Conclusion

Our systems supply target information at inference time and reach 85.22 and 91.78 F_{avg2}. Separate public comparisons support same-target demonstrations and retrieval, show a small target-dependent card effect, and find no benefit from the tested adapter. Target-shift methods should be selected on target-disjoint data. Model disagreement localises difficult rows but does not identify their correct labels.

Limitations

Track 1 uses external labelled examples for its evaluation target and is not directly comparable with task-data-only systems. We fix $k = 8$ and do not compare dense retrieval or demonstration order; ArabicStanceX also differs from Mawqif in annotation and class balance. The target-card study covers three targets, and its 0.39-point mean gain lacks a confidence interval. Track 2 lacks target-held-out component ablations, while count assignment assumes known label totals. The post-hoc rubric study covers one target and tests cross-corpus transfer. Hosted-model drift limits repeatability.

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